Query Optimisation

Mini Project Option-2

Utkarsh Jain (uj299) on December 19, 2024

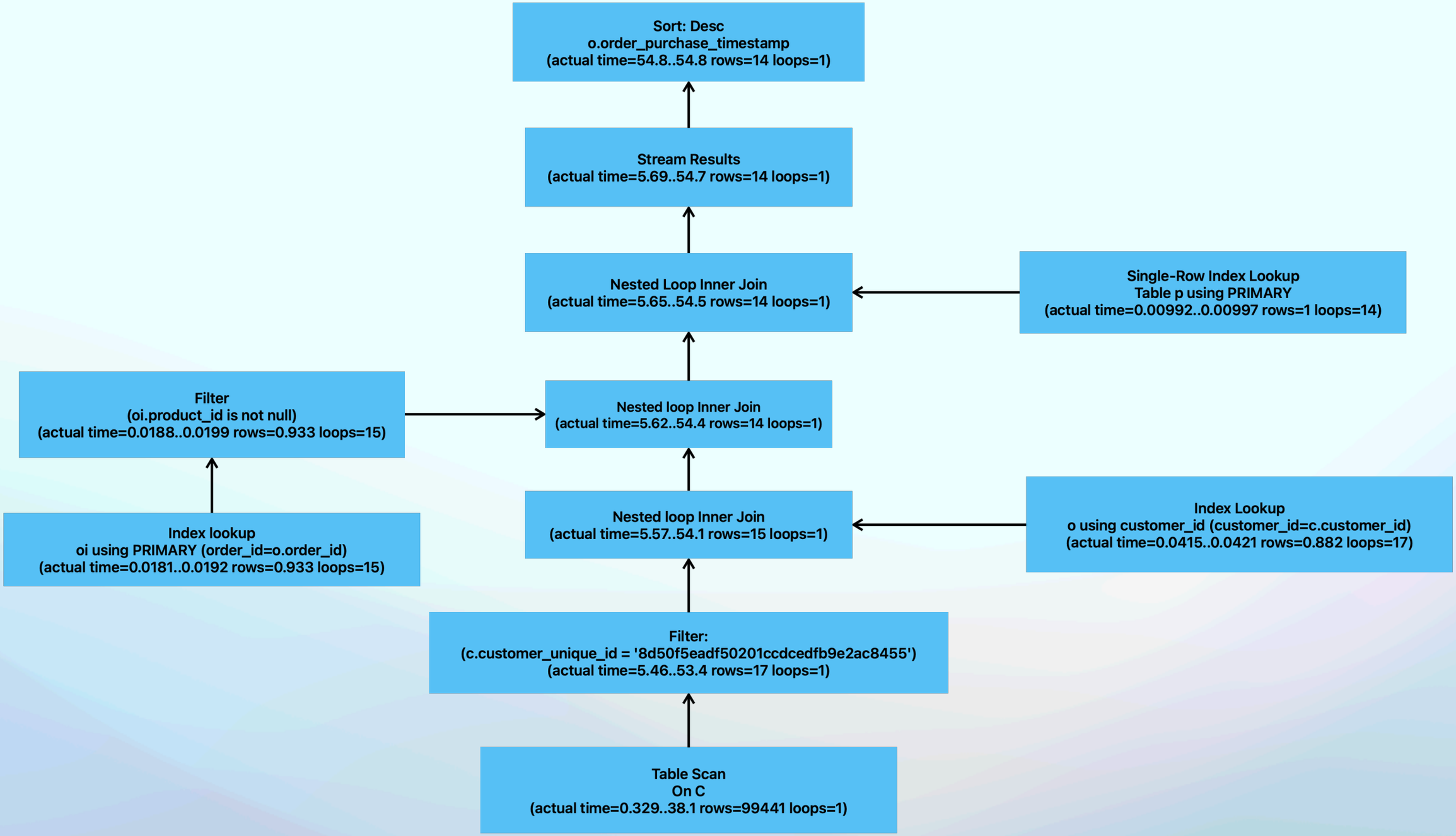
Current Performance

```
-> Sort: o.order_purchase_timestamp DESC  (actual time=54.8..54.8 rows=14 loops=1)
    -> Stream results  (cost=23843 rows=13765) (actual time=5.69..54.7 rows=14 loops=1)
        -> Nested loop inner join  (cost=23843 rows=13765) (actual time=5.65..54.5 rows=14 loops=1)
            -> Nested loop inner join  (cost=19026 rows=13765) (actual time=5.62..54.4 rows=14 loops=1)
                -> Nested loop inner join  (cost=14763 rows=11543) (actual time=5.57..54.1 rows=15 loops=1)
                    -> Filter: (c.customer_unique_id = '8d50f5eadf50201ccdcedfb9e2ac8455')  (cost=10723 rows=10442) (actual
time=5.46..53.4 rows=17 loops=1)
                        -> Table scan on c  (cost=10723 rows=104422) (actual time=0.329..38.1 rows=99441 loops=1)
                    -> Index lookup on o using customer_id (customer_id=c.customer_id)  (cost=0.276 rows=1.11) (actual
time=0.0415..0.0421 rows=0.882 loops=17)
                -> Filter: (oi.product_id is not null)  (cost=0.25 rows=1.19) (actual time=0.0188..0.0199 rows=0.933 loops=15)
                    -> Index lookup on oi using PRIMARY (order_id=o.order_id)  (cost=0.25 rows=1.19)
            -> Single-row index lookup on p using PRIMARY (product_id=oi.product_id)  (cost=0.25 rows=1) (actual
time=0.00992..0.00997 rows=1 loops=14)
```

On Running Explain Analyze

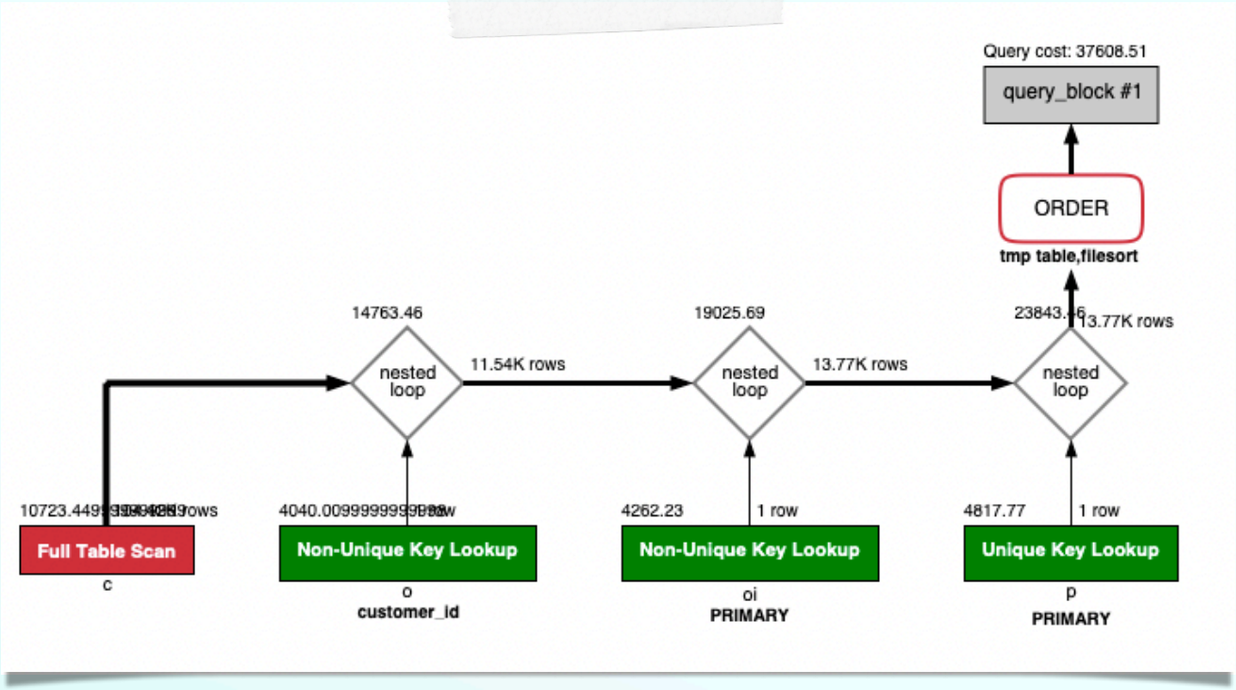
Explain Analyse

Visualised



Execution Plan

And Some Statistics



Timing (as measured at client side):
Execution time: 0:00:0.05537200

Timing (as measured by the server):
Execution time: 0:00:0.05521300
Table lock wait time: 0:00:0.00002900

Errors:
Had Errors: NO
Warnings: 0

Rows Processed:
Rows affected: 0
Rows sent to client: 14
Rows examined: 99498

Temporary Tables:
Temporary disk tables created: 0
Temporary tables created: 0

Joins per Type:
Full table scans (Select_scan): 1
Joins using table scans (Select_full_join): 0
Joins using range search (Select_full_range_join): 0
Joins with range checks (Select_range_check): 0
Joins using range (Select_range): 0

Sorting:
Sorted rows (Sort_rows): 14
Sort merge passes (Sort_merge_passes): 0
Sorts with ranges (Sort_range): 0
Sorts with table scans (Sort_scan): 1

Index Usage:
No Index used

Other Info:
Event Id: 126
Thread Id: 50

Improving Performance

Identifying Bottlenecks

Towards faster Execution

- Total Scan on the Customer Table scans about 99K rows!

Fixing Bottlenecks

Index Creation

- Creating 2 Indexes
  - olist_customers_dataset
  - olist_orders_dataset

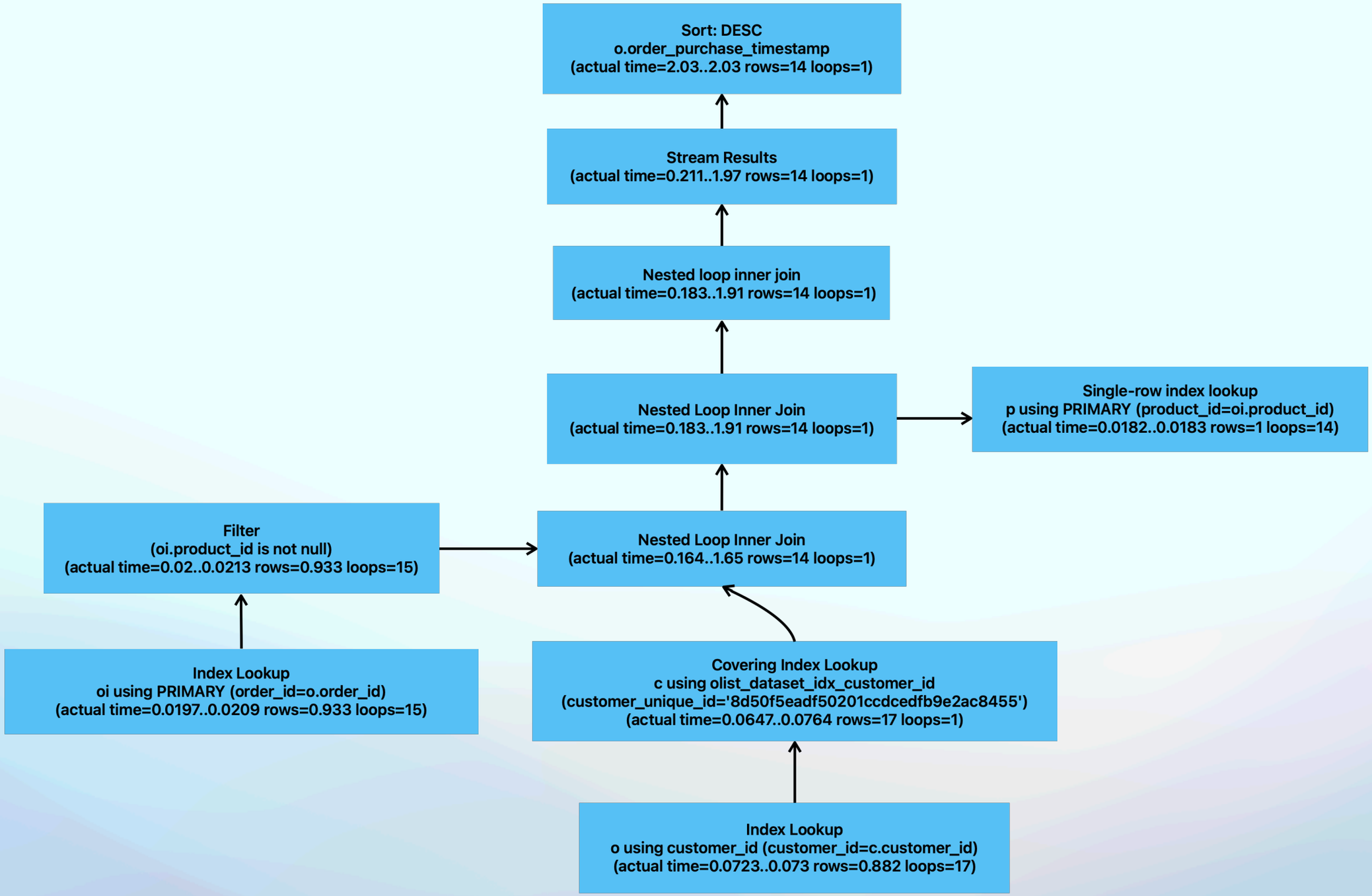
```
ALTER TABLE `olist_customers_dataset` ADD INDEX `olist_dataset_idx_customer_id` (`customer_unique_id`);
ALTER TABLE `olist_orders_dataset` ADD INDEX `olist_dataset_idx_timestamp` (`order_purchase_timestamp`);
```

```
-> Sort: o.order_purchase_timestamp DESC  (actual time=2.03..2.03 rows=14 loops=1)
    -> Stream results  (cost=24 rows=21.2) (actual time=0.211..1.97 rows=14 loops=1)
        -> Nested loop inner join  (cost=24 rows=21.2) (actual time=0.183..1.91 rows=14 loops=1)
            -> Nested loop inner join  (cost=16.6 rows=21.2) (actual time=0.164..1.65 rows=14 loops=1)
                -> Nested loop inner join  (cost=9.9 rows=18.4) (actual time=0.136..1.32 rows=15 loops=1)
                    -> Covering index lookup on c using olist_dataset_idx_customer_id
(customer_unique_id='8d50f5eadf50201ccdcedfb9e2ac8455')  (cost=3.46 rows=17) (actual time=0.0647..0.0764 rows=17
loops=1)
                    -> Index lookup on o using customer_id (customer_id=c.customer_id)  (cost=0.277 rows=1.08) (actual
time=0.0723..0.073 rows=0.882 loops=17)
                -> Filter: (oi.product_id is not null)  (cost=0.256 rows=1.15) (actual time=0.02..0.0213 rows=0.933 loops=15)
                    -> Index lookup on oi using PRIMARY (order_id=o.order_id)  (cost=0.256 rows=1.15) (actual time=0.0197..0.0209
rows=0.933 loops=15)
            -> Single-row index lookup on p using PRIMARY (product_id=oi.product_id)  (cost=0.255 rows=1) (actual
time=0.0182..0.0183 rows=1 loops=14)
```

On Running Explain Analyze

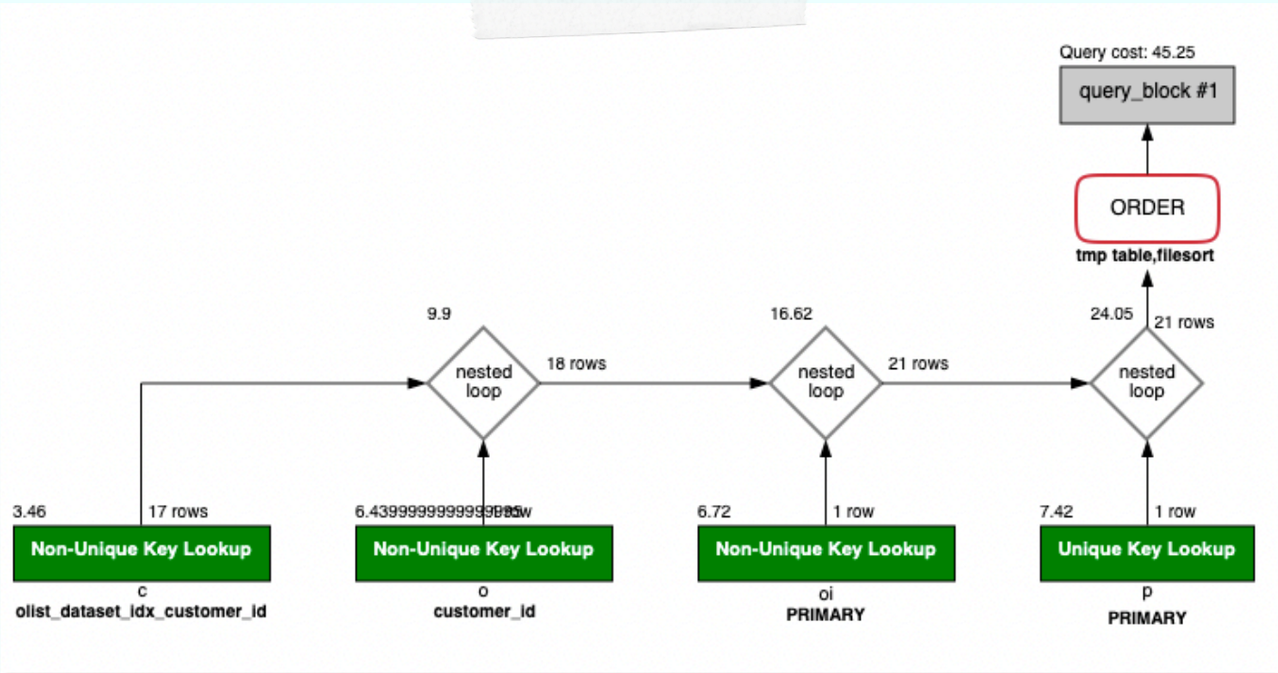
Explain Analyse

Visualised



Execution Plan

And Some Statistics



Timing (as measured at client side):
Execution time: 0:00:0.00508285

Timing (as measured by the server):
Execution time: 0:00:0.00494000
Table lock wait time: 0:00:0.00002700

Errors:
Had Errors: NO
Warnings: 0

Rows Processed:
Rows affected: 0
Rows sent to client: 14
Rows examined: 74

Temporary Tables:
Temporary disk tables created: 0
Temporary tables created: 0

Joins per Type:
Full table scans (Select_scan): 0
Joins using table scans (Select_full_join): 0
Joins using range search (Select_full_range_join): 0
Joins with range checks (Select_range_check): 0
Joins using range (Select_range): 0

Sorting:
Sorted rows (Sort_rows): 14
Sort merge passes (Sort_merge_passes): 0
Sorts with ranges (Sort_range): 0
Sorts with table scans (Sort_scan): 1

Index Usage:
At least one Index was used

Other Info:
Event Id: 140
Thread Id: 50

Comparision

Maybe it's not the thief of joy

	Old Query	New Query
Time (in milliseconds)	~54.8	~2
Rows Affected	99498	74

Fin.